

Reactive Photon Field Model (RPFM)

Model Description

This model provides a circuit-theory-inspired representation of photon physics using simple electromagnetic parameters.

It reformulates photon behavior in terms of effective circuit analogies such as capacitance, inductance, and vector potential, enabling a more intuitive and computationally accessible framework for analysis.

The approach opens a new perspective for studying complex physical concepts—such as photon energy and effective mass—through simplified parameter mappings rather than full field-theoretic formalisms.

This framework also supports practical interpretation through laboratory-scale signal mapping, where voltage is defined as:

$$V_{\text{map}} = A_0 c$$

using vector potential amplitude A_0 and the speed of light c as the scaling bridge between field representation and measurable electrical signals.

This voltage mapping provides an opportunity for model validation, allowing theoretical predictions to be compared directly with measurable laboratory or simulation-based voltage signals.

Project Scope

This project focuses on analyzing photon and electromagnetic field behavior using circuit-theory parameters.

The development is structured in stages:

- Stage 1: Construction and validation of the core computational model.
- Stage 2: Extension into an interactive simulation system using PyQt5 for visualization and parameter control.

Research Collaboration Intent

This project (RPFM) is developed as an open computational framework exploring photon wavelength–field interactions using reactive (LC-inspired) modeling analogies.

It is intended as a foundational simulation platform that can be extended, validated, and formalized under academic supervision for potential research applications.

The goal is to support further development in computational electromagnetics, wave modeling, and physics-based simulation systems through collaborative research.

Theory Introduction of RPFM Model

In standard physics, electromagnetic radiation is described as a propagating field solution of Maxwell’s equations, where energy is distributed continuously in space and time.

In contrast, the RPFM model proposes a structural decomposition of the photon (or electromagnetic quantum) into two coupled components:

- an inductive (magnetic) component
- a capacitive (electric) component

This approach interprets the photon not only as a propagating wave, but as a confined energy structure with localized field interaction volumes.

Core Concept

The total energy is expressed in analogy with oscillatory electrical systems:

$$mc^2 = hf \sim \frac{1}{2} \epsilon_0 E^2 AL + \frac{1}{2} \frac{B^2}{\mu_0} AL \quad (1)$$

and is reformulated as a combination of electric and magnetic energy storage terms.

$$U_t = U_b + U_e = \frac{1}{2} \epsilon_0 E^2 AL + \frac{1}{2} \frac{B^2}{\mu_0} AL$$

$$\text{Energy density } U = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 B^2$$

And $E=cB$ and $c^2 = 1/\epsilon_0 \mu_0$ and $B^2 / \mu_0 = \epsilon_0 E^2$ then

$$mc^2 = hf \sim AL/2 (\epsilon_0 E^2 + B^2/\mu_0) = \epsilon_0 E^2 AL = B^2/\mu_0 AL$$

$$mc^2 = hf \sim \epsilon_0 E^2 AL \quad \dots\dots\dots(2) \text{ Capacitive}$$

((hypothesis/model assumption))

$$mc^2 = hf \sim B^2/\mu_0 AL \quad \dots\dots\dots(3) \text{ inductive}$$

((hypothesis/model assumption))

Hypothesis:

Electric and magnetic field energy densities are assumed symmetric within a localized mode, and their combined energy scale is approximated as $\sim hf$.

Capacitive Interpretation

The electric-field contribution is modeled as:

$$mc^2 = hf \sim \epsilon_0 E^2 AL \text{ (hypothesis/model assumption)} \quad (2)$$

representing an effective capacitive energy density distributed over a finite spatial volume.

Inductive Interpretation

The magnetic-field contribution is modeled as:

$$mc^2 = hf = B^2 / \mu_0 A L \quad (3)$$

representing an effective inductive energy density associated with current-like field structure.

Physical Interpretation :

In this framework, the photon is treated as a coupled electromagnetic oscillator where:

- electric field energy behaves like a capacitive storage system
- magnetic field energy behaves like an inductive storage system
- geometry defines effective interaction volume

This leads to a unified structural interpretation of mass-energy in terms of field confinement.

In this model, the photon field is described using two effective geometric scales.
Longitudinal scale:

$$L_{eff} \sim \lambda$$

Transverse scale:

$$A_{eff} \sim \text{effective cross-sectional area of the field}$$

- Physical interpretation :
 - The longitudinal scale represents the propagation direction of the wave (wavelength scale).
 - The transverse scale represents the spatial spread of the field perpendicular to propagation.
- Volume decomposition

The effective field volume is defined as:

$$V = A_{eff} L_{eff}$$

- Key relation

$$L_{eff} \sim \lambda = 2\pi R$$

Vector potential :

A_0 is the organizing field variable from which the entire model is reconstructed.

$$\mathbf{B} = \nabla \times \mathbf{A}$$

$$A_0 \sim B_0 R$$

$$E_0 \sim \omega A_0$$

$$A_0 \sim \omega E_0$$

$$A_0 = B R = B \lambda / 2\pi \quad \text{and} \quad V = E R = E A_0 / B \quad \Rightarrow \quad V = A_0 c$$

- A_0 = field potential amplitude (carrier of wave information)
- R = geometric confinement scale
- ω = temporal oscillation scale
- E, B = derived field strengths

Lab Voltage Mapping Model :

In this framework, the voltage measured in laboratory or simulation settings is modeled as a direct scaling of the electromagnetic vector potential amplitude:

$$V_{\text{mapped}} = A_0 c \quad (4)$$

This model defines an effective mapping between field representation and measurable electrical voltage:

A_0 : vector potential amplitude of the electromagnetic wave

c : propagation speed (free-space scaling factor)

V_{mapped} : equivalent lab-observed voltage signal

This mapping treats the vector potential as the fundamental field descriptor, where the observed voltage is a scaled projection of wave propagation dynamics into a measurable electrical quantity.

Purpose in the model

- Converts simulated field amplitudes into measurable lab voltage signals
- Provides a direct amplitude-to-voltage calibration rule
- Enables comparison between theoretical wave models and experimental data

Project Equation :

Geometric constrains

$L_{eff} = 2 \pi R$ and $L_{eff} = \lambda$,
 then $R = L_{eff}/2 \pi = \lambda/2 \pi = 1/k$
 k is wave number.

$A_{eff} L_{eff} = V_{eff}$, effective volume

Capacitive part Model

For capacitive We have $m_{eff} c^2 = hf \sim \epsilon_0 E^2 V_{eff} = \epsilon_0 E^2 A_{eff} L_{eff}$

$$V_{eff} = A_{eff} L_{eff} \text{ and } L_{eff} = \lambda = 2 \pi R$$

in this project we have to areas forms

$$\begin{aligned} A_{eff} &= 4 \pi R^2 && \text{sphere} \\ A_{eff} &= \pi R^2 && \text{disk} \end{aligned}$$

$$1. A_{eff} = 4 \pi R^2$$

$$m_{eff} c^2 = hf \sim \epsilon_0 E^2 V_{eff} = \epsilon_0 E^2 A_{eff} L_{eff} = C_{eff} E^2 L_{eff}^2$$

And $E = B c$ and $A_{eff} = 4 \pi R^2$ we get

$$m_{eff} c^2 = hf = \epsilon_0 B^2 c^2 4 \pi R^2 L_{eff} = \epsilon_0 B^2 c^2 A_{eff} L_{eff}$$

$$m_{eff} c^2 = hf = (\epsilon_0 A_{eff}/L_{eff}) B^2 L_{eff}^2 c^2 \quad (5)$$

$C_{eff} = (\epsilon_0 A_{eff}/L_{eff})$,,effective capacitance.

$L_{eff} = \lambda$ and $A_{eff} = 4 \pi R^2$ then

$$C_{eff} = (\epsilon_0 4 \pi R^2/\lambda) = \epsilon_0 4 \pi \lambda^2/4 \pi^2 \lambda = \epsilon_0 \lambda/\pi$$

$$2. A_{eff} = \pi R^2$$

$$C_{eff} = (\epsilon_0 \pi R^2/\lambda) = \epsilon_0 \pi \lambda^2/4 \pi^2 \lambda = \epsilon_0 \lambda/4 \pi \quad (6)$$

Then energy equation became

$$m_{\text{eff}} c^2 = hf \sim C_{\text{eff}} B^2 L_{\text{eff}}^2 c^2 = C_{\text{eff}} B^2 4 \pi^2 R^2 c^2$$

$$B^2 R^2 = A_0^2 = B^2 \lambda^2 / 4 \pi^2 \quad \text{where } A_0 \text{ is vector potential .}$$

$$m_{\text{eff}} c^2 = hf = C_{\text{eff}} B^2 \lambda^2 c^2, \text{dividing by } c^2$$

$$m_{\text{eff}} = hf / c^2 = C_{\text{eff}} B^2 \lambda^2 = 4 \pi^2 C_{\text{eff}} A_0^2 \quad (7)$$

$$m_{\text{eff}} \propto C_{\text{eff}} A_0^2$$

mass contribution is represented through a capacitive term driven by vector potential amplitude squared. we can also obtain A_0 expression from m_{eff} . Equation

$$A_0 \propto 1/2 \pi \sqrt{\frac{m_{\text{eff}}}{C_{\text{eff}}}}.$$

In this framework, A_0 acts as a field amplitude parameter that link effective capacitance and effective mass scaling, enabling reconstruction of one from the other.

Mode-Dependent Effective Mass (capacitive General Form)

$$m_{\text{eff}}(n) = 4\pi^2 C_{\text{eff}}(n) (A_0(n))^2 \quad (8)$$

This expression defines the effective mass as a mode-dependent capacitive field quantity, where:

n : mode index (spatial, spectral, or harmonic mode).

$C_{\text{eff}}(n)$: effective capacitance of the field mode.

$A_0(n)$: vector potential amplitude of the mode.

$4\pi^2$: geometric normalization factor from circular/spherical mode structure.

Capacitive momentum :

$$m_{\text{eff}} c = hf / c = C_{\text{eff}} B^2 \lambda^2 = 4 \pi^2 c C_{\text{eff}} A_0^2$$

$$\text{For } A_{\text{eff}} = \pi R^2 \quad m_{\text{eff}} c = h/\lambda = 4 \pi^2 c C_{\text{eff}} A_0^2$$

$$m_{\text{eff}} c = h/\lambda = 4 \pi^2 c \epsilon_0 \lambda / 4 \pi A_0^2 = \pi c \epsilon_0 \lambda A_0^2$$

$$m_{eff} c = \pi \lambda A_0^2 / \mu_0 \quad c = \pi \lambda A_0^2 / Z_0 \quad (9)$$

$$\text{For } A_{eff} = \pi R^2 \quad m_{eff} c = h/\lambda = 4 \pi^2 c \quad C_{eff} A_0^2$$

$$m_{eff} c = h/\lambda = 4 \pi^2 c \quad \epsilon_0 \lambda / \pi \quad A_0^2$$

$$m_{eff} c = h/\lambda = 4 \pi \lambda A_0^2 / Z_0 \quad (10)$$

Gaussian curvature :

$$\text{For spherical } K_{eff} = 4\pi / A_{eff} \quad K_{eff} = k^2 \quad ,,, k \text{ is wave number}$$

$$\text{For disk } K_{eff} = \pi / A_{eff} \quad K_{eff} = k^2$$

PART 2: Inductive part Model

1. Fundamental Inductive Energy :

$$m_{eff} c^2 = hf = B^2 / \mu_0 \quad A_{eff} \quad L_{eff}$$

2. Geometric Constraints

$$L_{eff} = \lambda = 2 \pi R$$

3. Magnetic Field from Current

$$B = \mu_0 I / L_{eff} \quad (11)$$

$$B^2 = \mu_0^2 I^2 / L_{eff}^2$$

4. Substitution into Energy

$$m_{eff} c^2 = hf = B^2 / \mu_0 \quad A_{eff} \quad L_{eff} = (\mu_0^2 I^2 / \mu_0 L_{eff}^2) A_{eff} L_{eff}$$

$$m_{eff} c^2 = hf = \mu_0 I^2 A_{eff} / L_{eff} = (\mu_0 A_{eff} / L_{eff}) I^2$$

$$\text{effective inductance} \quad \mathcal{L}_{eff} = \mu_0 A_{eff} / L_{eff} \quad (12)$$

$$m_{eff} c^2 = hf = (\mu_0 A_{eff} / L_{eff}) I^2 = \mathcal{L}_{eff} I^2 \quad (13)$$

dividing by c^2

$$m_{eff} = hf = (\mathcal{L}_{eff} I^2) / c^2 = \mu_0 A_{eff} / L_{eff} \quad (B^2 L_{eff}^2 / \mu_0^2) / c^2$$

$$m_{eff} = (\mathcal{L}_{eff} \quad 4 \pi^2 A_0^2) / Z_0^2 \quad (14)$$

The inductive effective mass is defined as the inductance-weighted magnetic energy of a vector-potential-driven photon field normalized by vacuum impedance..

We also obtain A_0 expression like below

$$A_0^2 = Z_0^2/4 \pi^2 (m_{eff} / \mathcal{L}_{eff}) \Rightarrow A_0 = Z_0/2 \pi \sqrt{\frac{m_{eff}}{\mathcal{L}_{eff}}} \quad (15) \quad ,,,A_0 \text{ acts as a}$$

bridge variable, allowing one effective quantity to be inferred from another.”

Geometry Substitution :

- $A_{eff} = \pi R^2$ „For a photon in a physical system (fiber, cavity, beam, quantum optics).
- $A_{eff} = 4 \pi R^2$ „For free-space spherical emission.

$$L_{eff} = \lambda = 2 \pi R \quad , \quad R = \lambda / 2 \pi$$

- $A_{eff} = \pi R^2 = \pi \lambda^2 / 4 \pi^2$ thus $A_{eff} = \lambda^2 / 4 \pi$,then
- $$\mathcal{L}_{eff} = \mu_0 A_{eff} / L_{eff} \quad , \quad \mathcal{L}_{eff} = \mu_0 \pi \lambda^2 / 4 \pi^2 \lambda \quad , \quad \mathcal{L}_{eff} = \mu_0 \lambda / 4 \pi \quad (16)$$

- **Effective mass :**

$$m_{eff} c^2 = hf = (\mu_0 A_{eff} / L_{eff}) I^2 = (\mu_0 \lambda / 4 \pi) I^2$$

- $A_{eff} = 4 \pi R^2$ in the same way

$$m_{eff} c^2 = hf = (\mu_0 A_{eff} / L_{eff}) I^2 = (\mu_0 \lambda / \pi) I^2$$

$$m_{eff} c^2 = hf = (\mu_0 \lambda / \pi) B^2 L_{eff}^2 / \mu_0^2$$

$$m_{eff} c^2 = hf = (\mu_0 \lambda / \pi) B^2 \lambda^2 / \mu_0^2 = (\lambda / \mu_0 \pi) B^2 \lambda^2$$

$$\lambda^2 = (2 \pi R)^2 \text{ thus}$$

$$m_{eff} c^2 = hf = (\lambda / \mu_0 \pi) B^2 4 \pi^2 R^2 = (4 \pi \lambda / \mu_0) B^2 R^2$$

$$B^2 R^2 = A_0^2 \quad \text{Thus} \quad m_{eff} c^2 = hf = 4 \pi \lambda A_0^2 / \mu_0$$

$$m_{eff} c^2 = hf = 4 \pi \lambda A_0^2 / \mu_0 = (4 \pi \lambda / \mu_0) A_0^2 \quad \text{divid by } c^2$$

$$m_{eff} = hf/c^2 = (4 \pi \lambda / \mu_0 c^2) A_0^2 = 4 \pi A_0^2 / Z_0 c$$

$$m_{eff} = hf/c^2 = 4 \pi \lambda A_0^2 / Z_0 c \quad ,,, \quad Z_0 \text{ is free space impedance .}$$

$$m_{eff} = hf/c^2 = (4 \pi \lambda / Z_0 c) A_0^2 \quad (17)$$

• $A_{\text{eff}} = \pi R^2 = A_{\text{eff}} = \lambda^2 / 4\pi$ in the same way will get

$$m_{eff}c = hf/c^2 = \pi \lambda A_0^2 / Z_0 c = (\pi \lambda / Z_0 c) A_0^2 \quad (18)$$

• **Inductive momentum :**

From equation (14) $m_{eff} = (\mathcal{L}_{\text{eff}} - 4\pi^2 A_0^2) / Z_0^2$ multiply by c

$$m_{eff}c = (\mathcal{L}_{\text{eff}} - 4\pi^2 A_0^2) / Z_0^2 = (\mathcal{L}_{\text{eff}} - 4\pi^2 A_0^2) c / Z_0^2$$

From equation (12)

$$m_{eff}c = (\mu_0 A_{\text{eff}} / L_{\text{eff}} - 4\pi^2 A_0^2) c / \mu_0^2 c^2 = ((A_{\text{eff}} / L_{\text{eff}}) - 4\pi^2 A_0^2) / Z_0 \quad (19)$$

$$m_{eff}c = h/\lambda = (4\pi \lambda / Z_0) A_0^2 \quad \text{for } A_{\text{eff}} = 4\pi R^2 \quad (20)$$

$$m_{eff}c = h/\lambda = (\pi \lambda / Z_0) A_0^2 \quad \text{for } A_{\text{eff}} = \pi R^2 \quad (21)$$

• **quantum action and E & flux :**

From momentum equations for this model

$$m_{eff}c = h/\lambda = \pi \lambda A_0^2 / Z_0 \quad \text{for } A_{\text{eff}} = \pi R^2$$

$$m_{eff}c = h/\lambda = (\pi \lambda / Z_0) A_0^2$$

$$h/\lambda = (\pi \lambda / Z_0) A_0^2 \quad \text{then re-arrange } Z_0 h = \pi \lambda^2 A_0^2$$

$$A_0 = B \lambda / 2\pi \quad \Rightarrow \quad A_0^2 = B^2 \lambda^2 / 4\pi^2$$

$$Z_0 h = \pi \lambda^2 B^2 \lambda^2 / 4\pi^2 = B^2 \lambda^4 / 4\pi$$

$$Z_0 h = B^2 \lambda^4 / 4\pi \quad \text{then} \quad h = B^2 \lambda^4 / 4\pi Z_0$$

in this model $h = B^2 \lambda^4 / 4\pi Z_0$ is a field-based equivalent scaling representation of quantum action.

$$B^2 \lambda^4 = 4\pi Z_0 h \quad \Rightarrow \quad B \lambda^2 = \sqrt{4\pi Z_0 h}$$

$$\text{For } A_{\text{eff}} = \pi R^2 = \lambda^2 / 4\pi^2 \quad \Rightarrow \quad \lambda^2 = 4\pi A_{\text{eff}}$$

$$\Rightarrow \quad B 4\pi A_{\text{eff}} = \sqrt{4\pi Z_0 h} \quad \Rightarrow \quad B A_{\text{eff}} = \sqrt{4\pi Z_0 h} / 4\pi$$

$$\Rightarrow \Phi_B = \sqrt{4\pi Z_0 h} / 4\pi = \sqrt{Z_0 h / 4\pi} \quad \text{weber} \quad (22)$$

$$\Phi_E = c \Phi_B \Rightarrow \Phi_E = c \sqrt{Z_0 h / 4\pi}$$

$$\text{For } A_{\text{eff}} = 4\pi R^2 \quad \Phi_B = \sqrt{\pi Z_0 h} / \pi = \sqrt{Z_0 h / \pi}$$

$$\text{And } \Phi_B = \Phi_E / c \Rightarrow \Phi_E / c = \sqrt{Z_0 h / \pi} \Rightarrow$$

$$\Rightarrow \Phi_E = c \sqrt{Z_0 h / \pi}$$

$$\Rightarrow \Phi_E = c \sqrt{Z_0 h / \pi} \quad \text{V.m} \quad (23)$$

Note: The magnetic flux is assumed to vanish under symmetric free-space mode conditions within the model framework.

mass scaling rate :

we have from equation(20) $m_{\text{eff}} c = h/\lambda = (4\pi\lambda/Z_0) A_0^2$ for $A_{\text{eff}} = 4\pi R^2$

$$\Rightarrow h/\lambda = (4\pi\lambda/Z_0) A_0^2 \Rightarrow h/\lambda^2 = (4\pi/Z_0) A_0^2 \quad (\text{kg/s})$$

And from equation(21) $m_{\text{eff}} c = h/\lambda = (\pi\lambda/Z_0) A_0^2$ $A_{\text{eff}} = \pi R^2$

$$\Rightarrow h/\lambda^2 = (\pi/Z_0) A_0^2 \quad (\text{kg/s}) \quad (24)$$

The quantity h/λ^2 represents a wavelength-dependent mass scaling rate in the model's spectral space.